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Aryabhatta Knowledge University, Patna

B.Tech 5th Semester Exam., 2017

Time: 3 hours

Code: 051513 ***Subject: Computer Network*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Answer any seven of the following questions: **[14]**

(a) (a) Write any two functions of the session layer in OSI model.

(b) (b) What is Hamming code?

(c) (c) What is the significance of the twisting in twisted-pair cable?

(d) (d) What do you mean by piggybacking?

(e) (e) What is the purpose of the jam signal in CSMA/CD?

(f) (f) What are the advantages of dividing an Ethernet LAN with a bridge?

(g) (g) What does the term 'best effort' mean in context with IP?

(h) (h) Why is there no need of CSMA/CD in a full-duplex switched Ethernet?

(i) (i) How does redundancy facilitate error detection?

Q.2 Solve both questions: **[7+7=14]**

(a) Describe the architecture and working of ring topology. What are its advantages and disadvantages?

(b) Compare and contrast between the OSI model and the TCP/IP model.

Q.3 Solve both questions: **[7+7=14]**

(a) Describe the working of fiber-optic cables. Briefly explain different propagation modes used in fiber-optic cables.

(b) Explain Go-Back-N ARQ as a sliding window mechanism for error control.

Q.4 Solve both questions: **[7+7=14]**

(a) Describe how the two-dimensional parity check is able to detect errors. Consider '1100101100110000110100101101' as the data that is to be sent over the network.

(b) Draw a flowchart and explain the working procedure of the pure ALOHA protocol.

Q.5 Solve this question: **[14]**

Explain in detail Fast Ethernet. Describe the structure of the physical layer and various implementation types at this layer.

Q.6 Solve this question: **[14]**

With the help of a neat and labelled diagram, explain the message format of ICMP. Explain various types of ICMP error messages and query messages.

Q.7 Solve both questions: **[7+7=14]**

(a) What do you mean by layer 2 switching? Explain in detail the functioning of layer 2 switches.

(b) Discuss the functions and the design issues of the network layer in the OSI model.

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Q.8 Solve both questions: **[7+7=14]**

(a) Describe bridge in terms of networking device. What are different types of network bridge?

(b) Differentiate between Static Routing and Dynamic Routing.

Q.9 Explain the following terms: **[7x2=14]**

(a) (a) Border Gateway Protocol

(b) (b) User Datagram Protocol

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